AarogyaVaani

A voice-first, offline patient intake tool for primary health centres

Hack for Social Cause | VBYLD 2027 | Theme: Healthcare, Wellbeing & Service Delivery | State focus: Tripura

1. Problem Statement

State: Tripura

At primary health centres in Tripura, a doctor often has only a few minutes with each patient. Many patients speak Bengali or Kokborok, and many rural women cannot read. A full history is rarely taken. In pregnancy, danger signs such as bleeding, severe headache or fits can be missed. AarogyaVaani is an offline, voice-led intake tool. It gathers a complete history before the consultation and warns the health team about danger signs.

Target beneficiaries

- Pregnant women and other patients at public primary health facilities, especially in rural and tribal areas.
- ASHA workers and ANMs, who do the first contact and most of the paperwork.
- Doctors at PHCs and CHCs, and hub doctors who consult by tele-consultation.

Current gaps

- **Very short consultations.** India's average primary care consultation is about 2 minutes [1].
- **Language and literacy.** Bengali is the mother tongue of 63.43% of people in Tripura and Kokborok of 23.97% [3][4]. Nearly 1 in 4 rural women aged 15 to 49 are not literate [2].
- **Too few doctors.** About a third of medical officer and specialist posts are vacant [2].
- **History is on paper.** Records are scattered, so each visit starts from zero, and a doctor joining by tele-consultation has no history to read [8].
- **Existing tools do not fit.** Registration systems capture only name and age. Health apps need a smartphone, internet and prior sign-up [8].

2. The Problem in Numbers

Fact	Figure	Source
Average primary care consultation in India	**About 2 minutes**	[1]
Tripura population living in rural areas / from Scheduled Tribes	**73.83% / 31.76%**	[2]
Mother tongue: Bengali / Kokborok	**63.43% / 23.97%**	[3] [4]
Rural women (15 to 49) who are literate	**76.9%, so about 1 in 4 are not**	[2]
Women with 4 or more antenatal care visits	**52.7% in the state, 49.1% in rural areas. Unakoti district 24.6%, North Tripura district 67.1%**	[2]
Women (15 to 49) who are anaemic	**67.2%, up from 54.5%**	[2]
Medical officer and specialist posts vacant	**33% (ANM posts: 45%). Public health staff per 10,000 people: 11 in place against 17 sanctioned**	[2]
Urban PHCs / tribal-area CHCs against need	**5 in place, 29 needed / 8 in place, 13 needed**	[2]
Institutional deliveries that happen in public facilities	**90.7%**	[2]

Latest state-level data found: Census 2011, NFHS-5 (2019 to 2021) and HMIS 2019-20, compiled by NHSRC (2021). For four or more antenatal visits, the records-based HMIS figure is 69.5%; NFHS-5 is a survey.

What the numbers tell us

- Doctors are few, but ASHAs are in position almost everywhere: 7,077 against a target of 7,216 [2]. The first step of history-taking should move to the patient and the health worker, with the app as the guide.
- About nine in ten institutional deliveries happen in public facilities. A tool placed in public facilities reaches almost every woman who delivers in a facility.
- Language and literacy rule out typing and reading. Voice and pictures are needed. Kokborok is written in both Bengali and Roman script [5], which is one more reason to rely on voice and icons.

3. The Solution

AarogyaVaani runs on an Android phone or tablet kept at the health centre. The patient, or an ASHA or ANM on the patient's behalf, answers a guided interview by speaking or by tapping pictures. The app asks follow-up questions the way a doctor would, watches for danger signs, and builds a short summary for the doctor. The summary is read back to the patient in their own language.

The app works without internet and sends data to the health record when the network returns. Nothing is recorded without the patient's agreement. The app never makes a diagnosis. The doctor reviews, edits and approves every summary.

Who it helps

- **The patient:** is understood in their own language, with no reading or typing.
- **The health worker (ASHA or ANM):** is guided step by step, with less paperwork.
- **The doctor, on site or remote:** reads a complete history in seconds.

How a visit works

1. **Choose language and give consent.** A short audio message explains what the app will do. The patient agrees by voice or by tapping.
2. **Answer the interview** by voice or touch. An ASHA or ANM can run it in assist mode.
3. **The app asks follow-up questions** and checks each answer for danger signs. An urgent sign sends an instant alert to the nurse.
4. **The app builds the summary** and reads it back to the patient.
5. **The doctor reads, edits and approves.** Saved data syncs when the network is available.

Languages

The first release supports Bengali and Hindi, with picture screens for every question. Kokborok voice is planned for a later release. We found public Kokborok speech datasets and early community experiments, but no ready voice service [10]. Until then, Kokborok-speaking patients use the picture screens with help from a health worker who speaks the language.

Where it fits

AarogyaVaani sits in front of eSanjeevani, the national telemedicine service. In eSanjeevani, a health worker at the local centre does a basic consultation first and then connects the patient to a doctor at a hub [6]. Our summary gives that hub doctor the history before the call starts.

Safeguards

- The doctor approves every summary. The app gives no diagnosis.
- Danger-sign checks use fixed medical rules built from standard lists of pregnancy danger signs [7][9]. A doctor or senior nurse should review the final list.
- The app reads back what it heard so the patient can correct mistakes.
- Consent follows India's Digital Personal Data Protection Act 2023. The patient can refuse or withdraw at any time.

4. Features

MVP means the feature is in the working demo. **Roadmap** means it comes later.

No.	Feature	What it does	Phase
1	**Language and consent**	The patient picks a language. A short audio message explains what the app will do with their answers. They agree by voice or by tapping, and can refuse or withdraw later. *Example: A patient hears the explanation in Bengali and taps "I agree".*	MVP
2	**Voice-first intake**	Every question is spoken aloud and every answer can be spoken. Nothing needs to be read or typed. *Example: "How many months pregnant are you?" The patient answers aloud and the app records it.*	MVP (Bengali, Hindi)
3	**Touch and icon answers**	Any question can be answered by tapping big pictures: faces for yes and no, a body outline to show where it hurts, faces for pain levels. *Example: The patient taps the stomach on a body picture.*	MVP
4	**Adaptive follow-up questions**	The next question depends on the last answer, like a doctor's questions. When a problem is mentioned, the app asks when it started, what it feels like and what makes it better or worse. *Example: "Headache" leads to "Since when?" and "Is your vision blurred?"*	MVP
5	**Danger-sign alerts**	Fixed medical rules check every answer. If a serious sign appears, the patient is marked urgent and the nurse gets an instant alert. *Example: Bleeding in pregnancy shows a red alert.*	MVP
6	**Maternal and antenatal module**	A question set for pregnancy: months of pregnancy, earlier pregnancies, check-ups done, iron tablets, and the main danger signs (heavy bleeding, severe headache or blurred vision, swelling of face or hands, fits, less baby movement, high fever, severe stomach pain, repeated vomiting). *Example: "Is the baby moving as much as usual?" A "no" is flagged at once.*	MVP (flagship)
7	**Health worker assist mode**	An ASHA or ANM runs the same screens for patients who cannot use the device, such as the very old, the very unwell or first-time phone users. *Example: An ASHA sits beside an elderly patient and taps the answers.*	MVP
8	**Doctor summary**	A short, structured summary: main complaint, story of the problem, past illness, medicines, allergies and danger signs. The doctor edits, approves or rejects it. The same summary is read back to the patient. *Example: The doctor reads a half-page summary instead of asking for four minutes.*	MVP
9	**Offline use and auto-sync**	Everything runs on the device. Summaries are stored safely and sent to the health record when the network returns. *Example: A hill-area centre with no signal still completes the full intake.*	MVP offline, sync on roadmap
10	**Data protection and cleanup**	Temporary session data, such as audio and answers in progress, is deleted once the summary is saved. Summaries waiting to sync stay protected on the device. *Example: After the doctor approves, the raw audio is gone.*	MVP
11	**Document capture**	Photograph old prescriptions, lab reports or the mother and child card. The app reads them, picks out medicines and test results, and puts them in date order. *Example: Three crumpled prescriptions become one dated list.*	Roadmap
12	**Health record and tele-consult link**	Sends the approved summary to the patient's health record (ABHA, under the Ayushman Bharat Digital Mission) and to the hub doctor in a tele-consultation. *Example: The hub doctor opens the summary before the video call.*	Roadmap

Planned interface (API) for each feature

This is a design list. No technology is named. The same calls run on the device when the app is offline.

Feature	Planned calls
1 Language and consent	`POST /consents` `DELETE /consents/{id}`
2 Voice-first intake	`POST /sessions` `POST /sessions/{id}/answers (voice answer)`
3 Touch and icon answers	`POST /sessions/{id}/answers (touch answer, same call)`
4 Adaptive follow-up questions	`GET /sessions/{id}/next-question`
5 Danger-sign alerts	`GET /sessions/{id}/alerts` `POST /alerts/{id}/acknowledge`
6 Maternal module	`POST /sessions (module = maternal)`
7 Assist mode	`POST /sessions (mode = assisted)`

8 Doctor summary	`GET /sessions/{id}/summary`
	`PUT /sessions/{id}/summary (edit)`
	`POST /sessions/{id}/summary/approve`
	`GET /sessions/{id}/summary/audio (read-back)`

9 Offline use and auto-sync	`POST /sync (send queued summaries)` `GET /sync/status`
10 Data protection and cleanup	`DELETE /sessions/{id}/temp-data`

11 Document capture (roadmap)	`POST /documents`
	`GET /documents/{id}/extracted`
	`GET /patients/{id}/timeline`

12 Record and tele-consult link (roadmap)	`POST /records/push` `POST /teleconsult/handoff`

5. Impact and Scalability

How we will measure impact

We do not claim results yet. We will set numeric targets after a small pilot at one primary health centre. These are the measures.

Measure	What it shows
Time for a first-time patient to finish the intake	Whether it is usable without training
Share of patients who finish without help	Whether it works for low-literacy users
Share of summary fields filled	How complete the history is
Danger signs flagged, compared with a nurse checklist	Safety: missed signs and false alarms
Time from urgent flag to nurse attention	Speed of response
Time the doctor takes to read the summary	Time saved, against a consultation of about 2 minutes
Doctor edits per summary	Accuracy of the summary

Scalability

- One design fits every level of care. Tripura has 107 rural PHCs, 22 CHCs and 965 sub-centres [2], and the same intake runs at each.
- It uses the health workforce already in place: 7,077 ASHAs [2]. No new staff are needed.
- New modules (child health, chronic disease follow-up) and new languages are added as question sets and language packs.

- The model can be adapted for other states by changing the language and question sets.
- The code will be open source, as the hackathon requires.

SDG alignment: SDG 3 (Good Health and Well-being) and SDG 10 (Reduced Inequalities).

6. MVP and Roadmap

What we build first, so it can be shown in a live demo and a 3 to 5 minute video.

MVP

- One use case: maternal and antenatal intake, with about 10 to 15 core questions.
- Bengali and Hindi, with picture screens for every question.
- Voice and touch answers, plus assist mode for the health worker.
- Danger-sign rules for a small, well-chosen set of red flags, with an instant nurse alert.
- A structured summary on a simple doctor's screen, with edit and approve.
- Fully offline local storage and an audio consent screen.

Roadmap

- Kokborok voice support.
- Document capture and reading.
- Auto-sync, health record (ABHA) link and tele-consult hand-off.
- General adult intake and chronic disease follow-up.

7. Sources

[1] Irving G, et al. International variations in primary care physician consultation time: a systematic review of 67 countries. BMJ Open 2017;7:e017902.

[2] National Health Systems Resource Centre (NHSRC). Health Dossier 2021: Reflections on Key Health Indicators, Tripura. Uses Census 2011, NFHS-5, HMIS 2019-20, Rural Health Statistics 2019-20 and the HRH report 2020.

[3] Census of India 2011, population by mother tongue (Table C-16), as summarised in Languages of Tripura.

[4] Directorate of Kokborok and Other Minority Languages, Government of Tripura. Kokborok language note (880,537 speakers, 23.97%, Census 2011).

[5] IAS Gyan. Kokborok language (Bengali and Roman scripts both in use).

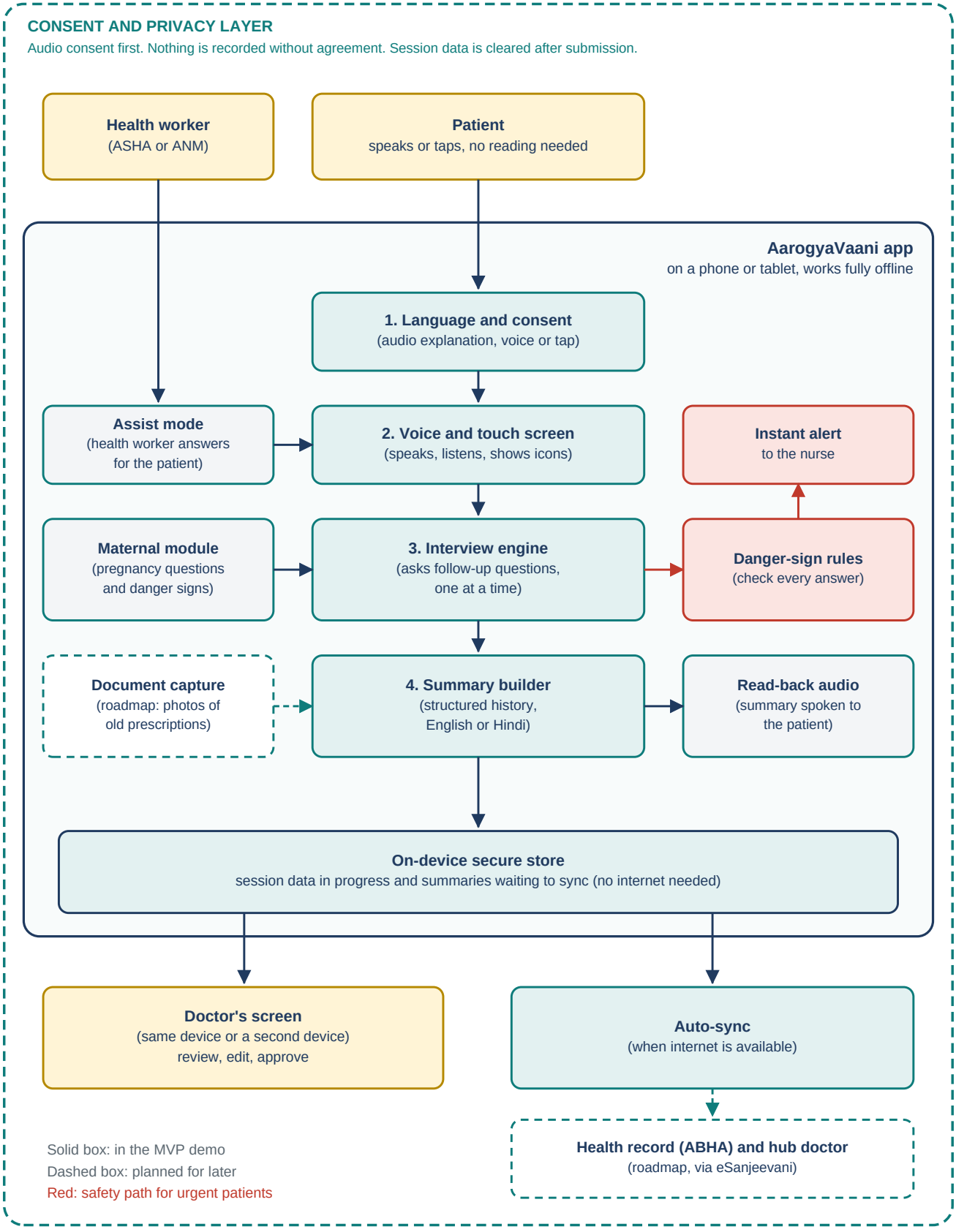
[6] IMPRI. eSanjeevani: India's National Telemedicine Initiative; Ministry of Health and Family Welfare, eSanjeevani National Telemedicine Service.

[7] Vikaspedia (Government of India portal). Danger signs during the course of a pregnancy.

[8] Smart India Hackathon 2026, problem statement SIH26047: Patient Case-Taking Software.

[9] Annals of Global Health 2024;90(1):31. Obstetric danger signs based on the WHO Guide for Essential Practice (Table 1).

[10] AIKosh (Bhashini): SpeeD-TB Kokborok speech dataset; Kathbath and Fleurs Bengali speech datasets (AI4Bharat).

8. Architecture Diagram

How AarogyaVaani fits together. The patient or health worker talks to the app. Everything runs on the device. Data leaves only when the network is available.